

External Review of the gdTAI Training Program

Independent methodological review of data preparation, model/algorithm selection, training, and evaluation of the gdTAI γ δT-cell classifier

- **Review date:** 2026-08-06
 - **Reviewer role:** external methodological reviewer (commissioned audit)
 - **Project:** gdTAI v1-v4 γ δT classifier series, trained on the 5,128,904-cell integrated T/NK atlas
 - **Reviewed model of record:** gdTAI v3.0 (`gdta_i_v3`, promoted default, `balanced_fixed_0p936` operating mode), with v3 Round 12 as validated high-purity fallback, v2.0 as reference release, v4.0 as experimental
 - **Constraint accepted for this review:** no additional datasets can be obtained; all recommendations are executable with the data already in the project.
-

1. Executive summary

gdTAI is an unusually well-instrumented single-cell classifier project: artifacts are SHA-256-pinned, every candidate lane emits versioned reports, negative-control cohorts are screened under explicit frozen-profile protocols, and failure modes are audited rather than hidden. The promoted v3.0 model reaches recall ≈ 0.86 at specificity ≈ 0.999 on the one truly independent cohort (BALF_BLOOD_COPD), which is a credible result.

The review nevertheless finds that **the evidentiary basis for the headline performance numbers is weaker than the numbers suggest**, for three structural reasons:

1. **Leakage-prone evaluation design.** Train/tune splits are made at the cell level, so donors, libraries, and batches straddle train and tune; feature lists were selected on the full object before splitting; and the full-atlas metrics used for final model selection are computed on the same cells used to report performance. The one clean whole-dataset holdout (GSE144469 in v2) was moved into training in v3.
2. **Adaptive, post-hoc model selection.** Roughly 38 v3 candidate rounds plus at least six separate candidate lanes (v1/v2 strategies, NK-guard, multi-guard, transcriptome-gate, annotation-specific cascade, v4) were screened against the same validation cohorts, with selection criteria tightened after results were inspected (target recall 0.75 during training $\rightarrow$ 0.80 guardrail later), a promoted model (Round 12) subsequently replaced by a retrospectively renamed Round 14, and the “final-test-only” external cohort used inside the promotion and selection rules. This is a textbook garden-of-forking-paths pattern; the external cohort is no longer untouched.
3. **No replicated external positive-truth validation.** The 8-cohort extension screen (758,135 cells) contains zero gdT-gold/silver cells and can only measure false-positive rates. A single external cohort (948 gdT positives) carries the entire external recall claim, and its primary evaluation excludes ambiguous cells.

Overall verdict: the gdTAI v3.0 artifact is usable for atlas-internal, high-specificity γ δT candidate calling, and the project’s negative-control discipline is genuinely good practice.

However, the promoted model’s reported recall/precision should be treated as optimistic internal estimates rather than validated external performance, the model registry and selection history need consolidation, and the program should shift from “more rounds” to **honest uncertainty quantification, prespecification, calibration, and robustness analysis** — all achievable without new data.

Highest-priority recommendations (full list in §6):

- **R1.** Freeze the evaluation protocol: declare BALF_BLOOD_COPD and the two sealed extension holdouts (GSE315928, GSE121636) as never-touch test sets; stop adaptive iteration against them; document the Round 12→Round 14 replacement openly.
- **R2.** Quantify split optimism with leave-one-dataset-out cross-validation across the labeled sources and donor/library-cluster bootstrap confidence intervals on every headline metric.
- **R3.** Add calibration analysis (reliability curves, isotonic recalibration on tune) and prevalence-linked operating-point guidance to the reporting.
- **R6.** Make per-dataset metrics the primary reporting frame (pooled metrics currently mask source-level FP fractions ranging from ~0.5% to ~24%).

2. Scope, materials, and method

The review covered the complete gdTAI lineage and its evaluation machinery:

Layer	Primary materials reviewed
Label construction	workflows/gdtai/ run_gdt_prediction_package_evaluation.py (gold/ silver rules, GSE144469 boolean repair, sublibrary evidence); Integrated_dataset/tables/ gdT_prediction/ground_truth_*.csv
Training	workflows/gdtai/ run_gdt_deg_tcr_classifier_training.py (v1/v2), run_gdt_gse144469_holdout_tcrgene_classifier.py (v2 holdout), run_gdtai_v3_trdc_nk_guard_classifier.py (v3), run_gdtai_v4_tcell_gate_classifier.py (v4), run_gdtai_nkguard_classifier.py, run_gdtai_multiguard_classifier.py, run_gdtai_transcriptome_gate_cascade.py, run_gdtai_annotation_specific_cascade.py
Evaluation	workflows/gdtai/ compare_gdtai_v3_round12_vs_round14.py, compare_frozen_gdtai_profiles.py, compare_gdtai_latest_vs_v2_high_purity.py, failure-audit scripts; reports and tables under Integrated_dataset/logs/tables/figures/ gdT_prediction/**

Layer	Primary materials reviewed
Governance	configs/models/gdtai/model_registry.csv, configs/models/gdtai/README.md, configs/models/ gdtai/extension_evaluation.json, gdT_prediction/ gdT_classifier_local_handoff_plan.md

Method: independent read-through of the executable scripts and their emitted tables/logs/reports; headline numbers cross-checked against source CSV/JSON artifacts (label counts, split sizes, filter counts, and Round 12/14 metrics were re-verified against `ground_truth_class_counts.csv`, `split_overall.csv`, `hard_negative_expression_filter_summary.csv`, and `gdtai_v3_round12_vs_round14_summary.json` during this review). No pipeline code, H5AD, model artifact, or registry entry was modified.

2.1 Ground-truth label system (as found)

Labels are built from TCR metadata in `run_gdt_prediction_package_evaluation.py`:

- **gdT_gold (57,776 cells, 1.13% of atlas):** every cell of the three sorted- $\gamma\delta$ T cohorts (GDT_2020AUG_woCOV 25,904; GDTlung2023july_7p 15,175; MalteGDT 7,800) **regardless of TCR evidence**, plus HRA005041/GSE144469 cells in gdTCR-sequenced sublibraries with paired TRG/TRD and no $\alpha\beta$ TCR (8,897 cells).
- **abT_gold (603,203 cells):** paired TRA/TRB, no $\gamma\delta$ TCR — but only in HRA005041 and GSE144469, the two `TCR_EVALUATION_SOURCES`.
- **gdT_silver (2,194 cells):** TRD-only, sensitivity analyses only.
- **unlabeled_or_ambiguous (4,465,731 cells, 87.1%):** everything else, including all TRAB-sequenced cells from the other ~60 datasets.

A sublibrary is flagged “gdTCR-sequenced” if **any** cell in it has $\gamma\delta$ TCR evidence (`build_sublibrary_labels`, script lines 469–506): in HRA005041 only 8 of 321 sublibraries qualify.

3. Strengths of the current process

Credit is due — several practices here exceed typical single-cell classifier work:

1. **Artifact integrity.** SHA-256 pinning of model artifacts and input H5ADs, verified before use (`compare_gdtai_v3_round12_vs_round14.py`:184-200); registry-stored hashes.
2. **Explicit negative-control discipline.** The extension screen is honest about what it can measure: “these studies used alpha-beta TCR assays and contributed 0 gdT-gold and 0 gdT-silver cells... this report estimates false-positive behavior and prediction burden only” (`compare_frozen_gdtai_profiles.py`:1737). Sealed holdouts (GSE315928, GSE121636) are registered in `extension_evaluation.json`.
3. **Failure-mode auditing as a first-class artifact.** Three separate failure audits (v3 TRDC/NK, annotation-specific TP/FN, GSE144469 holdout) quantify NK contamination, TRDC+TRDV— ambiguity, and per-tissue/per-annotation FN structure rather than burying it.
4. **Statistical texture.** Wilson intervals on rates, donor-cluster bootstrap in the extension screen, McNemar tests explicitly flagged as descriptive because of intra-donor

correlation, and prevalence-aware PPV tables (PPV $\approx$ 0.205 at 0.1% prevalence vs 0.948 at the 6.58% validation prevalence).

5. **Reproducible randomness.** Fixed seeds throughout (`RANDOM_SEED = 20260501`), offset RNG streams for splits and subsampling.
6. **Documented fallback policy.** Round 12 is retained as a validated high-purity fallback rather than silently deleted; v4 is explicitly marked experimental-not-promoted.

These strengths mean the recommendations below are about *locking in* rigor, not rebuilding the project.

4. Findings: data preparation and labeling

F1 (Critical) — Cell-level splits leak donor/library/batch structure into tuning

All three training scripts split **cells** within `(source_gse_id, label)` strata (v3: `split_by_source_label`, `run_gdtai_v3_trdc_nk_guard_classifier.py`:635-653; v1/v2: `make_gold_splits`, `run_gdt_deg_tcr_classifier_training.py`:737-827). Cells from the same donor, library, 10x channel, and batch therefore appear in both train and tune. Single-cell expression is strongly donor- and batch-correlated, so tune-set metrics — used for threshold and candidate selection — are optimistically biased. Only the v2 holdout script isolates an entire dataset (GSE144469) before training, and that dataset was moved into v3 train/tune (`GSE144469_primary_in_train_tune = 63,764 cells`, `v3_split_overall.csv`).

F2 (Critical) — Label-feature circularity through the TCR pipeline

Gold labels are TCR-metadata-derived, and the deployable feature space is dominated by TCR gene expression, which carries essentially the same information. The project's own "diagnostic" strategies prove recoverability: v1 strategy B (TCR metadata as features) reaches perfect validation precision/recall (`classifier_training/classifier_metrics_validation.csv`). This is not fatal — a classifier that reproduces TCR calls from RNA alone has real utility where VDJ data are absent — but it means reported metrics measure *TCR-pipeline reproduction fidelity*, bounded by the noisiness of RNA-level TCR detection, not biological $\gamma\delta$ T identification accuracy. Reports should state this framing explicitly.

F3 (Major) — Feature-selection leakage and a DEG feedback loop

- The v1/v2 DEG feature list was computed on the **full** object before splitting (`select_deg_features`, `run_gdt_deg_tcr_classifier_training.py`:399-436), so validation cells influenced feature choice.
- Those DEGs were defined by contrasting the previous classifier's TN/FN cells (`run_tn_fn_trd_gt0p1_deg_reproducibility.py`:263-275) — the next model's features are derived from the last model's errors, a closed feedback loop.
- Phase-4 module scores (themselves model outputs on the same object) enter v1/v2 as features.

F4 (Major) — Post-hoc label repair on a dataset later used for training

GSE144469 required a source-specific boolean repair because the upstream harmonizer treated literal "NA" strings as non-empty TRD calls (`build_corrected_tcr_evidence`,

`run_gdt_prediction_package_evaluation.py:414-466`; audit table
`tcr_evidence_correction_audit.csv`). v3 then trains on these post-hoc-repaired labels.

F5 (Major) — Sorted- $\gamma\delta$ T cohorts assumed 100% positive without TCR evidence

All 48,879 cells of the three sorted cohorts are labeled `gdt_gold` unconditionally (`gdt_gold_rule_sorted = sorted_gdt, make_truth_labels`). GDTlung2023july_7p (15,175 cells) is acknowledged as suboptimal library quality and excluded from some training lanes, yet remains in the global gold count and in sensitivity frames. Sorting purity is not verified by TCR within these cohorts.

F6 (Major) — Negative-set construction is unrepresentative in both directions

- **NK hard negatives are filtered to the rare TRDC+TRDV– pattern:** of 110,149 prefilter NK-annotated cells, 105,266 (95.6%) were discarded (`hard_negative_expression_filter_summary.csv`). The model never sees representative NK negatives, while 96% of retained NK negatives are exactly the ambiguous pattern.
- **Weak-TRDC negatives are pre-selected by the phase-4 TRD score** (top quartile), i.e. cells that already look $\gamma\delta$ -like are declared negative — easing the discrimination task and potentially suppressing recall on genuine $\gamma\delta$ T in that score region.
- **abT_gold negatives exist for only 2 of ~60 datasets** (HRA005041, GSE144469); paired-TCR cells elsewhere (e.g., GSE254249) are class-code 0 proxies, so the negative class is a mixture of true gold and unlabeled proxy.

F7 (Major) — 87% of the atlas is unlabeled and excluded from evaluation

4,465,731 of 5,128,904 cells are `unlabeled_or_ambiguous`. Atlas-wide FP burden is **extrapolated**: the paired-TCRAB call rate in TCR-sequenced sources is multiplied onto non-TCR-sequenced sources (`run_gdtai_v3_trdc_nk_guard_classifier.py:2091-2099`), assuming equal FP rates across sources with different chemistries, tissues, and disease states. The estimated FP fraction (0.0406 for Round 14) is therefore a model-based estimate, not a measurement, and its uncertainty is unquantified.

F8 (Minor) — No batch correction in the classifier feature space

Features are per-cell $\log_{10}$ (CP10K) values with no batch handling; cross-GSE chemistry and sequencing-depth differences enter the model directly. This is partially mitigated by the TCR-gene-focused feature set, but library-chemistry effects (5' vs 3', VDJ enrichment efficiency) are only discussed qualitatively.

5. Findings: model/algorithm selection, training, and evaluation

F9 (Critical) — Adaptive, multi-round selection against the same validation cohorts

The v3 lane alone enumerates ~38 candidate rounds (model families $\times$ post-hoc gates $\times$ threshold grids; `gdtai_v3_trdc_nk_guard_report.md` candidate table), alongside six further one-off lanes (v1/v2 strategies, NK-guard, multi-guard, transcriptome-gate cascade, annotation-specific cascade, v4). There is **no formal hyperparameter search** (all hyperparameters are

hard-coded single values), but the effective search over families, gates, and thresholds is large, and every branch is scored on the same tune/validation frames. The cascade wrappers add further post-hoc threshold grids (`run_gdtai_transcriptome_gate_cascade.py:512-518`; `run_gdtai_annotation_specific_cascade.py:251-261`). Without a prespecified primary candidate and a multiplicity-aware selection rule, the winner's measured advantage over runners-up is not interpretable at face value.

F10 (Critical) — Selection criteria changed after results were seen

- The v3 script defaults to `--target-recall 0.80`, but the reported run used **0.75** (internal tune table, `target_recall=0.75`). Round 12's full-atlas recall (0.7560) passes 0.75 and fails 0.80.
- Round 12 was promoted as gdTAI v3.0 (`gdtai_v3_trdc_nk_guard_summary.json`); a later comparison applied a **0.80 recall guardrail**, found Round 12 failing (0.7560 vs required ≥ 0.80), and promoted Round 14 (`compare_gdtai_v3_round12_vs_round14.py:831-914`).
- The promoted "Round 14" was originally **v3_round20** in the training script's own numbering and was retrospectively renamed by the comparison script; the registry README warns of a stale Round 12 override at the canonical path (`configs/models/gdtai/README.md:11-15`). The 0.80 floor's timing — after Round 12 was known to fall short — is exactly the kind of criterion drift prespecification exists to prevent.

F11 (Critical) — The external cohort is used for selection, not only final testing

The v3 script states the independent external H5AD is "used only for final testing" (`run_gdtai_v3_trdc_nk_guard_classifier.py:7-8`), yet: (a) v3 promotion gates compare external NK FP, TRDC+TRDV— burden, F1, and cytotoxic recall against v2-high-purity (`promotion_acceptance`, lines 2522–2575); and (b) the Round 12-vs-14 selection maximizes **mean F1 across full-atlas, atlas-held-out, and the independent external evaluation** (`compare_gdtai_v3_round12_vs_round14.py:891-909`). BALF_BLOOD_COPD has therefore directly influenced which model is the default. It remains independent of *fitting*, but it is no longer an untouched test set, and future "external validation" claims must account for this. Compounding this, the external primary evaluation excludes ambiguous cells (`stress_ambiguous` group; `external_truth`, lines 1676–1717), so the 38,995-cell primary frame (948 positives) is cleaner than real deployment.

F12 (Major) — Full-atlas metrics serve both selection and reporting

`select_candidate_from_full_atlas` (`run_gdtai_v3_trdc_nk_guard_classifier.py:2188-2255`) picks the final candidate using full-atlas recall and the extrapolated FP fraction — the same full-atlas numbers then reported as performance. The atlas includes train/tune cells, so these are training-set-adjacent metrics.

F13 (Major) — Single, unreplicated external positive-truth cohort; pooled metrics mask heterogeneity

- The only external recall/precision measurement rests on one cohort of 948 positives; the 8-cohort extension screen contains **zero** gdT-gold/silver cells (`gdtai_extension_evaluation_summary.md`) and supports FPR claims only.
- Per-dataset behavior varies dramatically: known-FP fraction by source ranges from $\sim 0.5\%$ (GSE287301) to $\sim 24\%$ (GSE243572) for the v2 selected model (`gse144469_holdout_tcrgene_report.md` per-source table); annotation-specific FN rates

reach 62.5% (colon tissue) and 100% (Treg-annotated cells) (`gdta_i_annotation_specific_tp_fn_audit_report.md`). Pooled headline metrics do not convey this deployment-relevant variance.

- No multiplicity correction is applied across the dozens of model × cohort × stratum comparisons.

F14 (Major) — No calibration analysis at the operating point

The promoted model applies a hard 0.936 threshold to a Platt-calibrated score, but no reliability curve or calibration error is reported at or around the operating point, and no decision-curve analysis links the threshold to a declared target prevalence — despite the project’s own PPV tables showing PPV swings from 0.205 to 0.948 across plausible prevalences. Users cannot currently choose an operating point rationally for their sample composition.

F15 (Minor) — Reproducibility and registry hygiene gaps

- No environment lock file (`pyproject.toml` declares only Python ≥ 3.10); LightGBM is an optional dependency absent from the reported run; exact package versions unpinned.
- The registry tracks four releases, but the many experimental variants and wrapper artifacts (`best_candidate_model.pkl`, `selected_model.pkl` in overlapping directories) are unregistered, and the Round 12/Round 14 override history is recoverable only by reading three files side by side.
- Deterministic seeds are set, but threaded histogram boosting and RNG-dependent negative subsampling make bitwise reruns fragile without a locked environment.

6. Recommendations (all executable without new data)

Ordered by priority. R1–R3 should be done before any further model iteration.

R1 — Prespecify and freeze the evaluation protocol. Write a one-page analysis plan that (a) declares BALF_BLOOD_COPD, GSE315928, and GSE121636 as never-touch test sets for any future model change; (b) freezes the selection rule and guardrails *before* the next comparison run; (c) documents the Round 12→Round 14 history, including the round-20 renaming and the 0.75→0.80 criterion change, in `DECISIONS.md`. From this point, any candidate evaluated against the sealed sets more than once is declared exploratory.

R2 — Quantify split optimism with grouped resampling. (a) Run leave-one-dataset-out cross-validation across the labeled sources (train on HRA005041 + sorted cohorts, test on GSE144469, and permutations) to get an honest cross-dataset generalization estimate; (b) recompute all headline metrics with donor/library-cluster bootstrap CIs (the extension screen already implements donor-cluster bootstrap — reuse it); (c) report a donor-level split sensitivity analysis for v3 to bound the optimism in the current tune numbers. Expected cost: a few training reruns of the already-scripted v3 lane.

R3 — Add calibration and prevalence-linked operating-point reporting. Reliability curves and expected calibration error for the promoted model on tune and on BALF_BLOOD_COPD; isotonic recalibration on the tune split if miscalibrated; decision-curve analysis across the prevalence scenarios already tabulated; a documented recommended operating point per declared prevalence band (screening vs confirmation use).

R4 — Stress-test the two load-bearing estimates. (a) Label-noise sensitivity: perturb gold labels at rates consistent with TCR-pipeline error (e.g., 1–5% symmetric flips, TRD-dropout simulation for silver→negative) and report metric stability; (b) FP-extrapolation sensitivity: recompute the atlas FP fraction under source-stratified and worst-case rate assumptions instead of the single equal-rate assumption, and report a plausible range rather than a point value.

R5 — Ship a three-zone output instead of forcing binary calls. Positive / negative / abstain (indeterminate) zones, with the TRDC+TRDV– quadrant routed to abstain by default; report the coverage-error tradeoff. This directly addresses the unresolved TRDC+TRDV– ambiguity (finding in all three failure audits) without pretending it is resolved.

R6 — Make per-dataset and per-stratum metrics the primary frame. Promote the existing per-source tables to the headline report; apply FDR control within each comparison family; treat pooled metrics as secondary. A model that is excellent on average but has a 24% FP fraction in one source is not uniformly deployable.

R7 — Consolidate governance artifacts. Register or delete the experimental variants; resolve the Round 12 override at the canonical path; export a locked environment (`conda env export`) and pin it in the model package; extend the existing `USAGE_PROTOCOL.md` model-card pattern to v3.0 with intended use, prevalence guidance, known failure modes, and the selection history.

R8 — Chemistry and dropout robustness suite. Stratify all metrics by 5' vs 3' chemistry and VDJ-enrichment status (metadata already in the registry); run gene-detection downsampling and GSE169246-style missing-feature stress tests on the frozen model to quantify robustness to assay differences — the extension screen already flags this cohort class but does not propagate its effect quantitatively.

R9 — Demonstrate added value over simple baselines on identical frames. Benchmark the promoted model against the TRD–TRAB module score, TRDC alone, and a small marker panel on exactly the same evaluation cells, with cluster-bootstrap CIs on the differences. If the margin over a 5-marker logistic rule is small, prefer the simpler deployable model.

R10 — Resolve TRDC+TRDV– using existing atlas evidence. Before any new data: examine TRG-family expression, cytotoxic programs, scVI cluster membership, and the existing annotation-evidence-review outputs for these cells across sources; classify them as NK-contaminant vs $\gamma\delta$ -lineage-consistent and encode the conclusion as an explicit policy (R5's abstain zone or a retraining decision).

7. Limitations of this review

- The review is code-and-artifact based; no models were retrained and no metrics recomputed beyond cross-checking emitted tables against their source CSV/JSON files (all spot-checked headline numbers matched).
- Judgments about criterion drift (F10) are based on file timestamps, defaults vs reported run parameters, and artifact naming history as committed; the reviewers did not interview the development team.

- “No more data” was accepted as a hard constraint; several issues (single external positive-truth cohort, zero-positive extension cohorts) are fundamentally only fixable with new gdTCR-sequenced data and are noted as residual risks rather than actionable findings.

Appendix A. Verified headline numbers (cross-checked 2026-08-06)

Quantity	Value	Source artifact
Atlas cells / gdT_gold / abT_gold / silver / unlabeled	5,128,904 / 57,776 / 603,203 / 2,194 / 4,465,731	Integrated_dataset/tables/gdT_prediction/ground_truth_class_counts.csv
v3 train/tune primary	505,640 / 126,410 cells (24,082 / 6,021 gdT_gold)	.../gdtai_v3_trdc_nk_guard/split_overall.csv
NK hard-negative discard	105,266 of 110,149 prefilter NK cells excluded (train)	.../hard_negative_expression_filter_summary.csv
Round 12 full-atlas precision / recall / F1 / est. FP frac	0.9823 / 0.7560 / 0.8544 / 0.0247	.../gdtai_v3_round12_vs_round14_summary.json
Round 14 full-atlas precision / recall / F1 / est. FP frac	0.9752 / 0.8056 / 0.8823 / 0.0406	same
Round 12 vs 14 atlas-held-out recall	0.6751 / 0.8530	same
Round 12 vs 14 external precision / recall / specificity	0.9690 / 0.8576 / 0.99932 vs 0.9509 / 0.8586 / 0.99890	same (948 positives)
Extension screen positive truth	0 gold, 0 silver across 758,135 cells (8 cohorts)	.../gdtai_extension_evaluation_summary.md
Prevalence-dependent PPV	0.205 @ 0.1% prevalence; 0.948 @ 6.58%	.../gse144469_holdout_tcrgene_report.md

Appendix B. Key file inventory

Training: workflows/gdtai/run_gdt_prediction_package_evaluation.py, run_gdt_deg_tcr_classifier_training.py, run_gdt_gse144469_holdout_tcrgene_classifier.py, run_gdtai_v3_trdc_nk_guard_classifier.py, run_gdtai_v4_tcell_gate_classifier.py, run_gdtai_nkguard_classifier.py, run_gdtai_multiguard_classifier.py, run_gdtai_transcriptome_gate_cascade.py, run_gdtai_annotation_specific_cascade.py. Evaluation: workflows/gdtai/compare_gdtai_v3_round12_vs_round14.py, compare_frozen_gdtai_profiles.py, compare_gdtai_latest_vs_v2_high_purity.py, run_gdtai_v3_trdc_nk_failure_audit.py, run_gdt_classifier_failure_mode_audit.py. Governance: configs/models/gdtai/model_registry.csv, configs/models/gdtai/README.md, configs/models/gdtai/extension_evaluation.json, gdT_prediction/gdT_classifier_local_handoff_plan.md. Reports/tables: Integrated_dataset/logs|tables|figures/gdT_prediction/**.